

Introduction to Materials Informatics (2026)

Skolkovo Institute of Science and Technology

Artem Dembitskiy, Researcher (Skoltech), Head of Data Research (Sber AI)

Dmitry Aksyonov, Assistant Professor (Skoltech)

Time: TBD

Location: TBD

Lecture #1: What is materials informatics about? + Course navigation

Goals

- Provide an understanding of what materials science and materials informatics are
- Describe the course content
- Help students decide if the course is right for them

Agenda

- Goals
- What is materials informatics?
 - Materials science
 - Trial-and-error paradigm
 - Experiment and theory
 - Materials informatics
 - Examples
- Course navigation
 - ILOs, HW, FP, assessment criteria

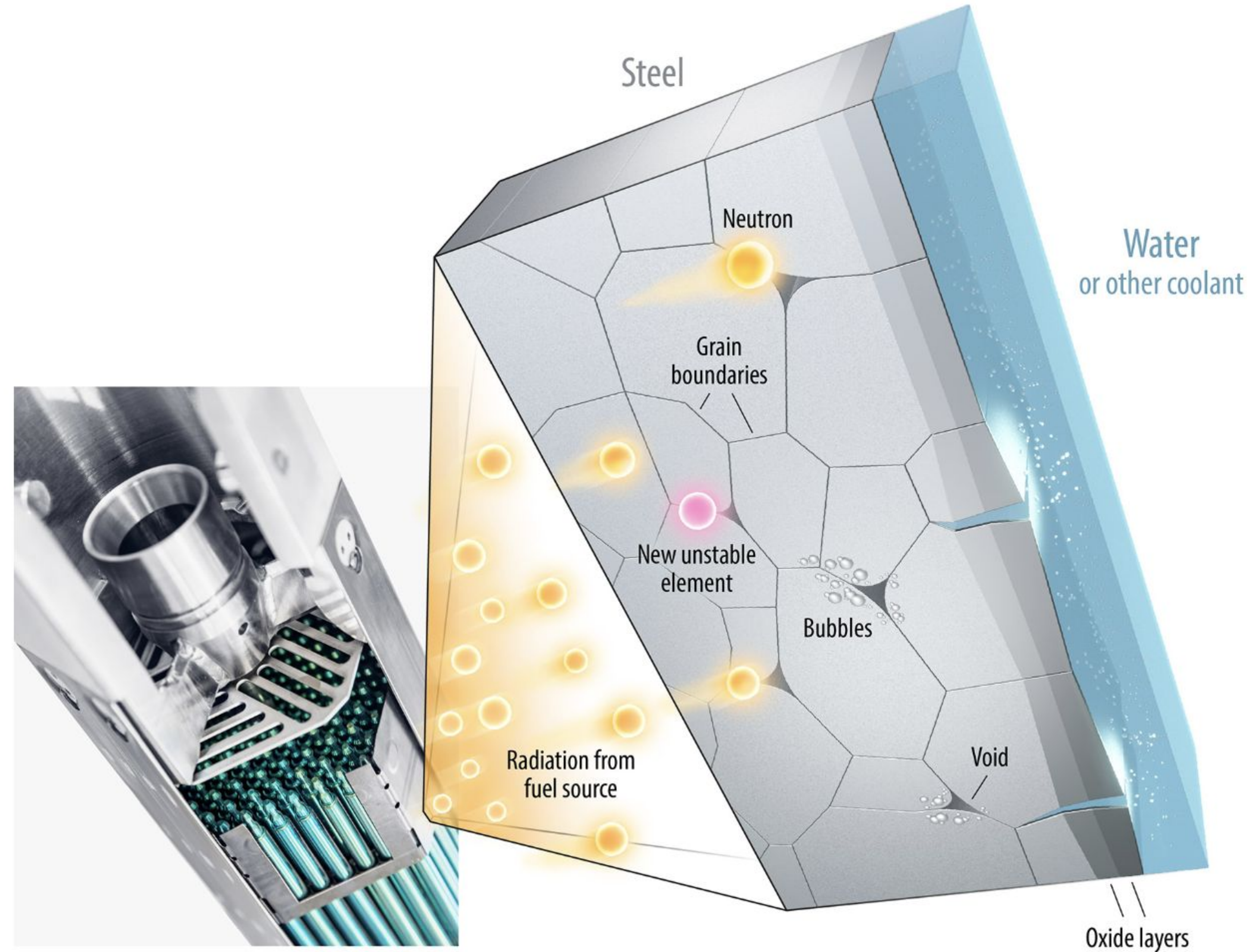
Part #1: Materials science and informatics

Materials science (and engineering)

- Studies the relationship between the structure and properties of a material
- Uses this knowledge to design and improve materials for practical applications

How to design/select?

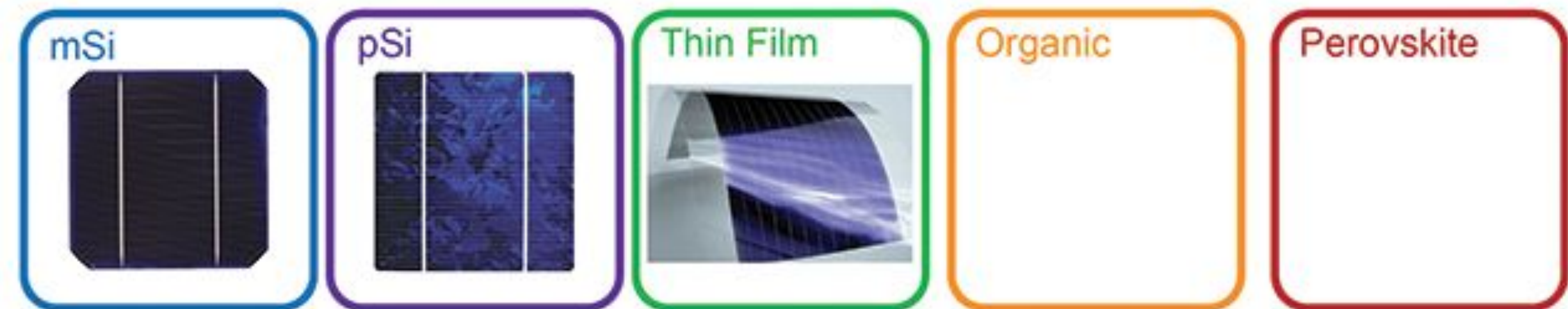
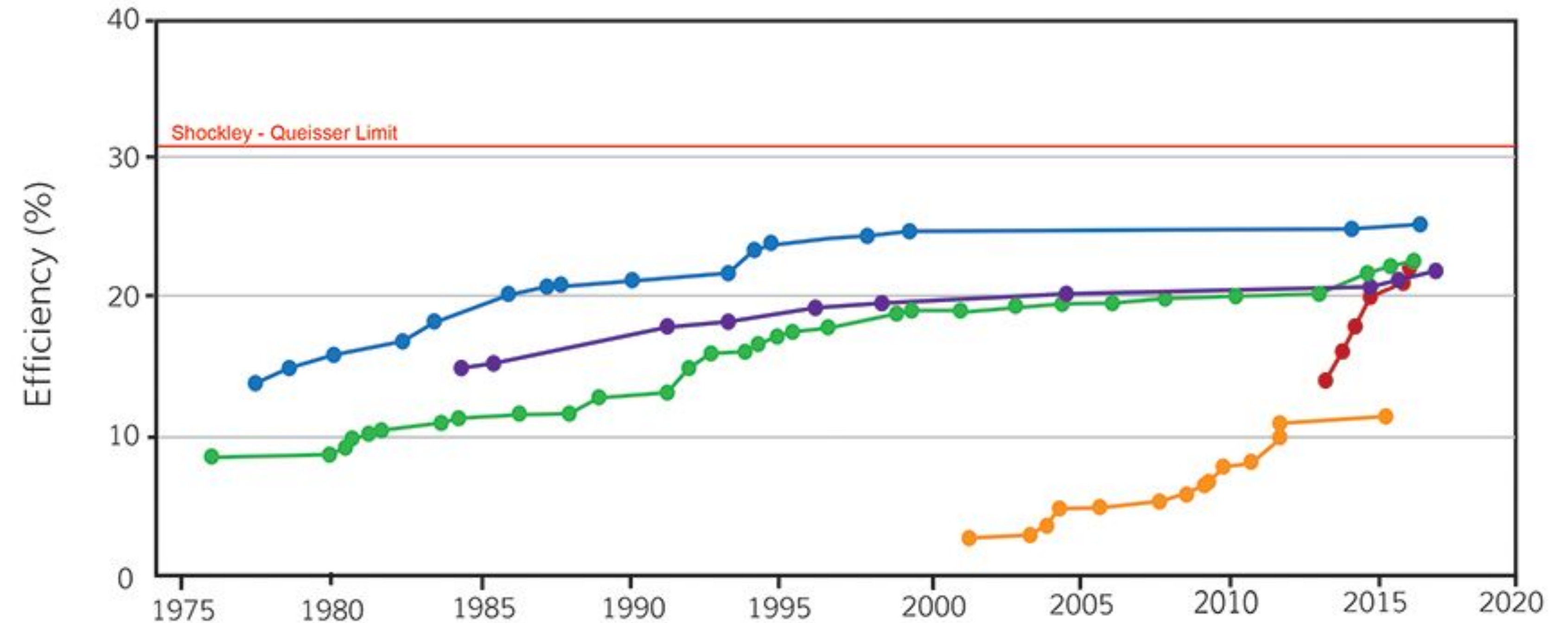
- Materials
- With specified properties
- Stable at given conditions



How to improve?

- Better props
- Less expensive
- Easier to process

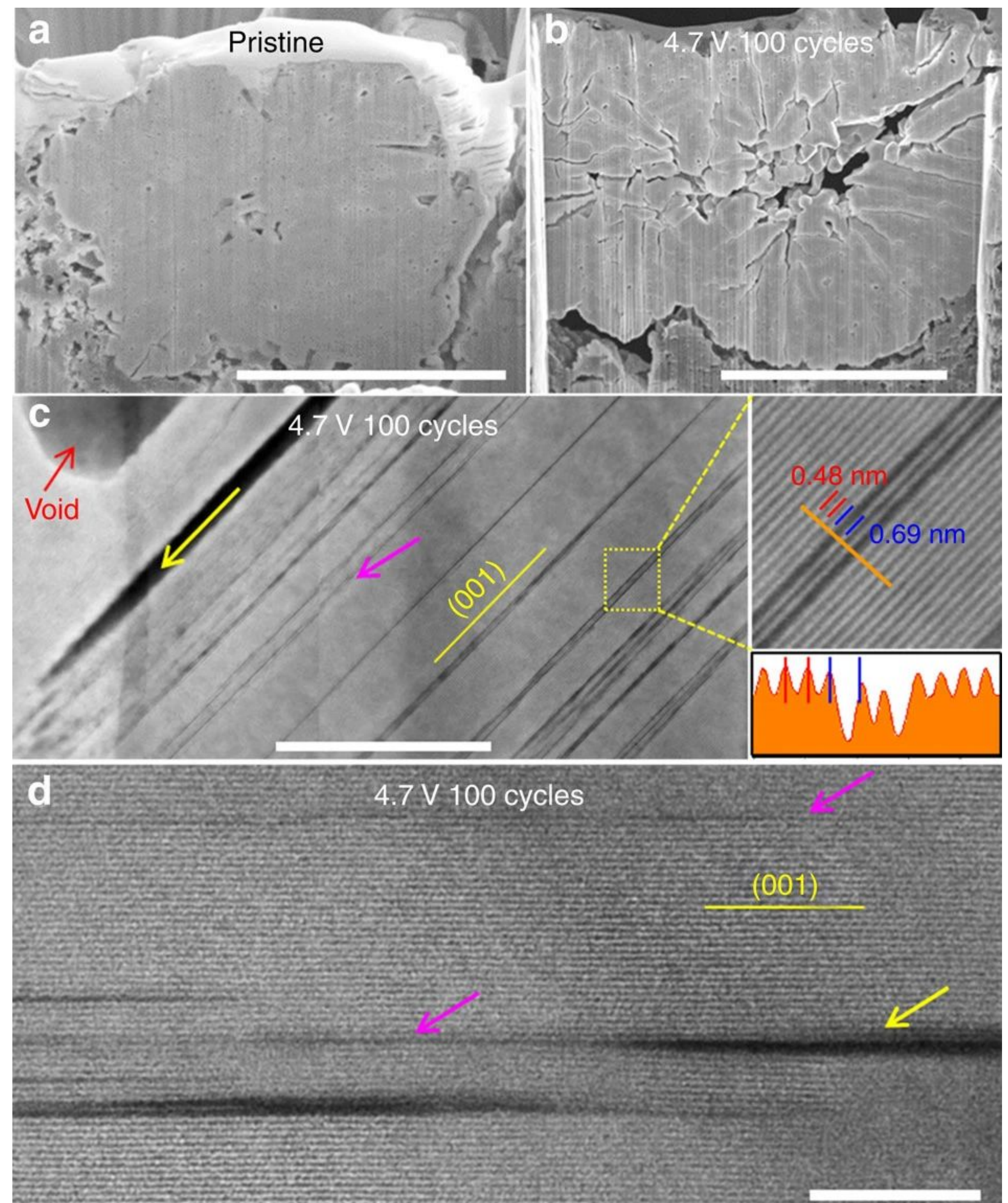
Lab Bench Record for Solar Cell Efficiency



Source: NREL

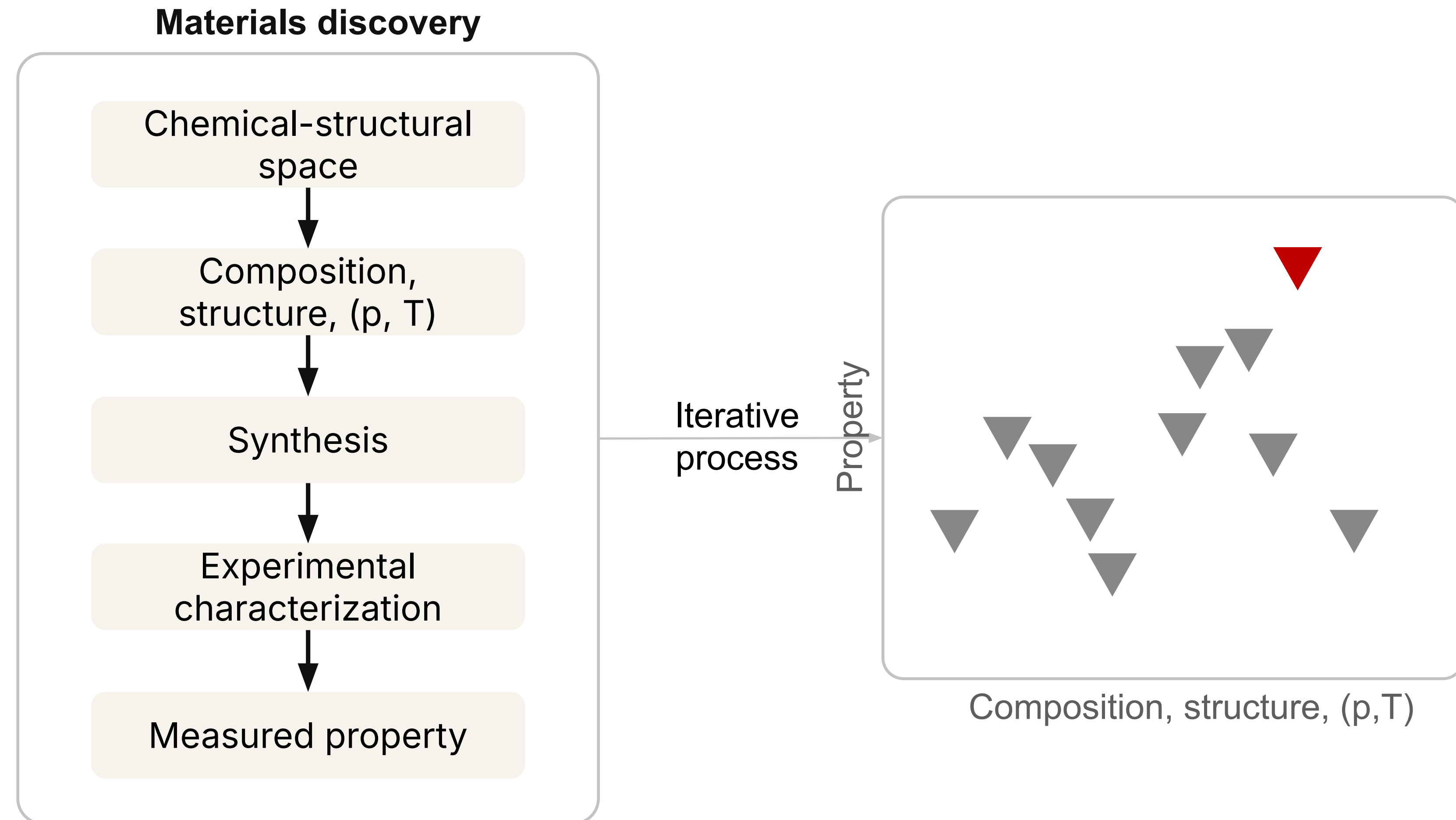
Why materials fails?

- Failure mechanism
- Can be fixed?



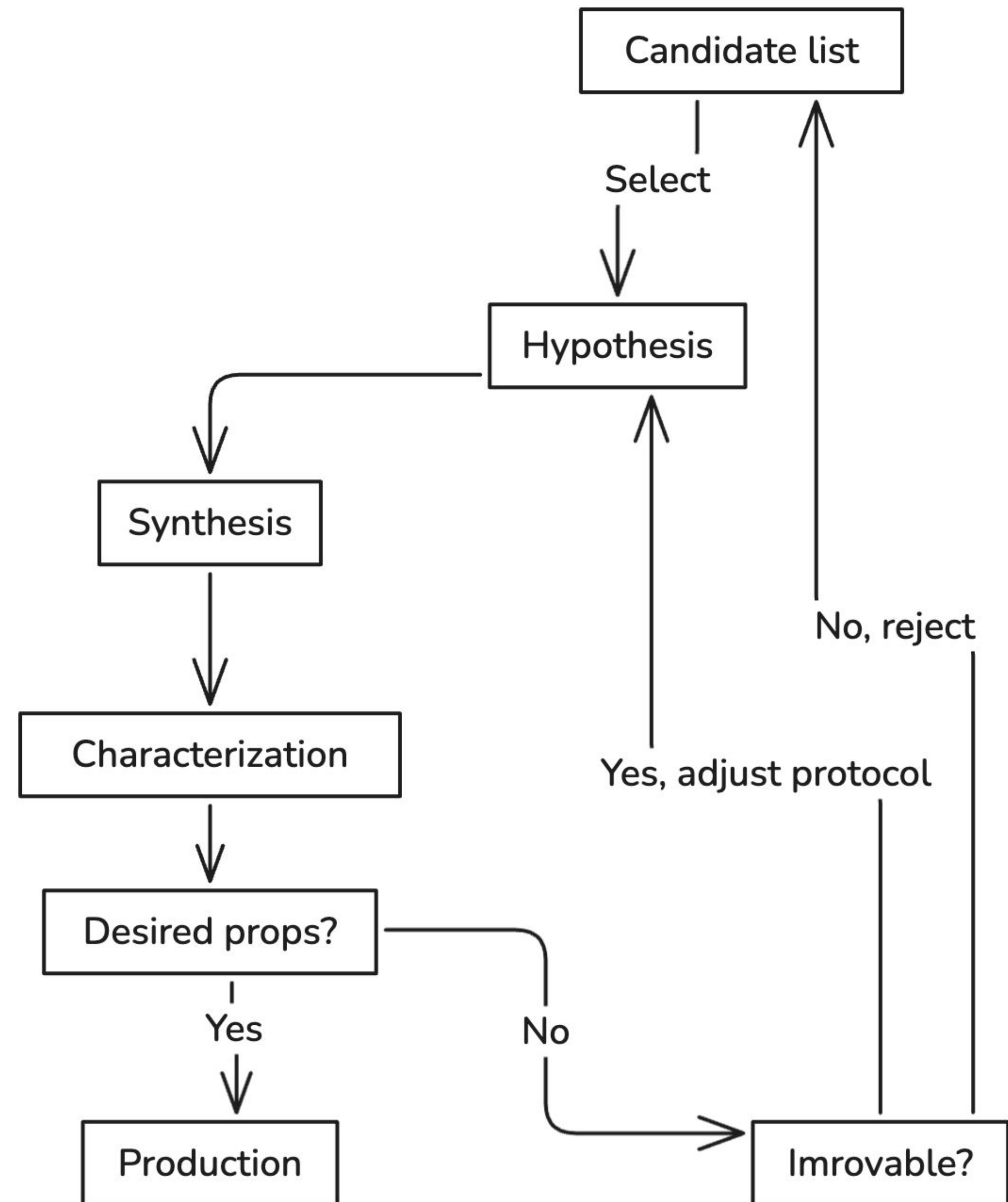
Trial-and-error paradigm

- Iterative process of searching for a material with desired properties



Trial-and-error in materials science

- The only way to reach the goal
- Takes ~10 years to design a material

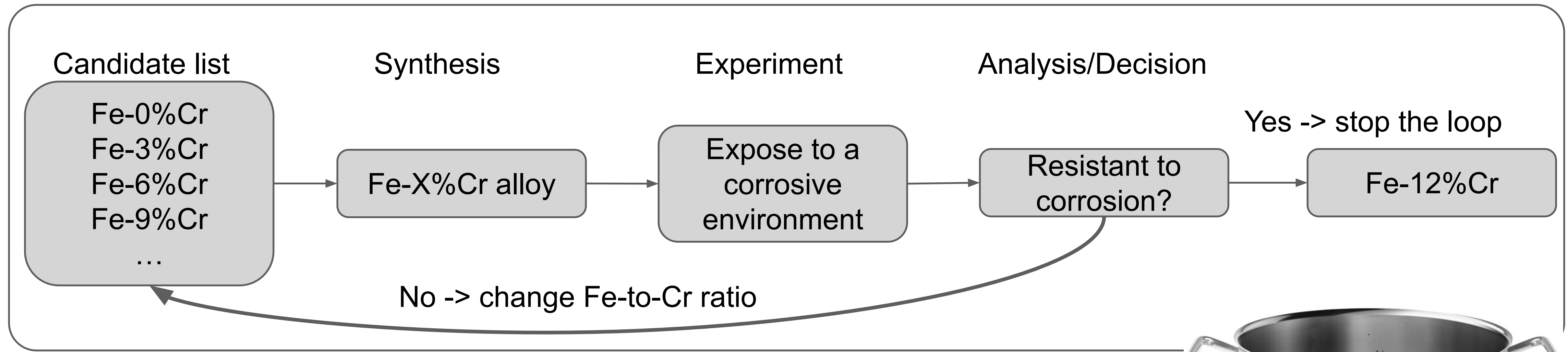


Stainless steels example

- 1798 - chromium discovery
- 1821 - iron-chromium alloys
- 1861 - the first patent on chromium steels
- 1911 - report on the relationship between the chromium content and corrosion resistance



Hypothesis: Fe-Cr alloys do not corrode*



*Oversimplified scheme



Bottleneck #1: Synthesis/Experimentation

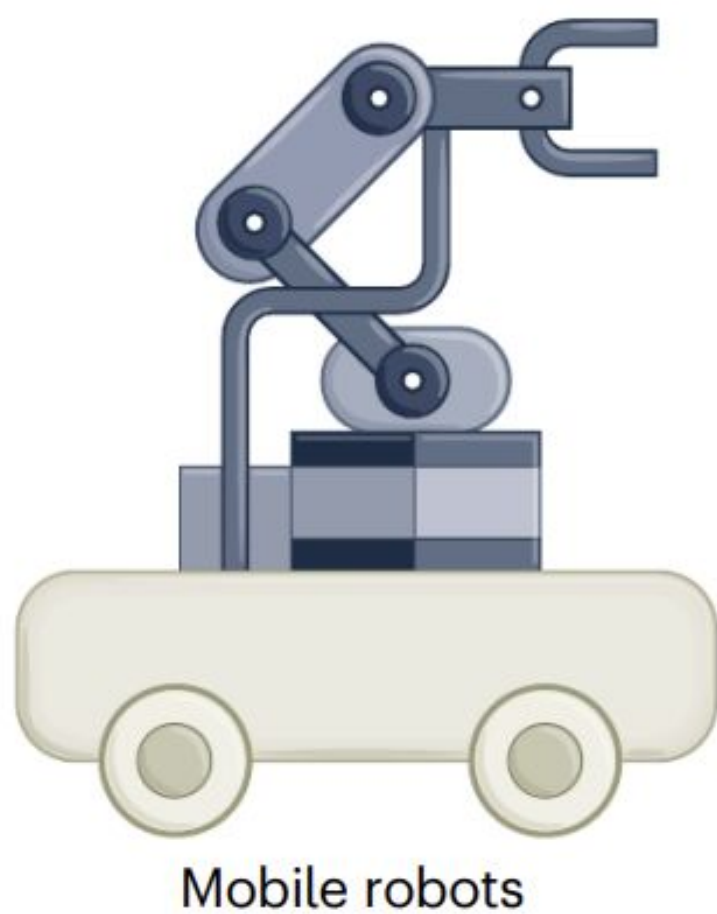
How to accelerate experimental studies?

- Optimize synthesis protocols
- Automate experiments, characterization
- Accelerate post-processing routines

Bottleneck #2: Size of a search space

How to navigate this space before stepping in a lab?

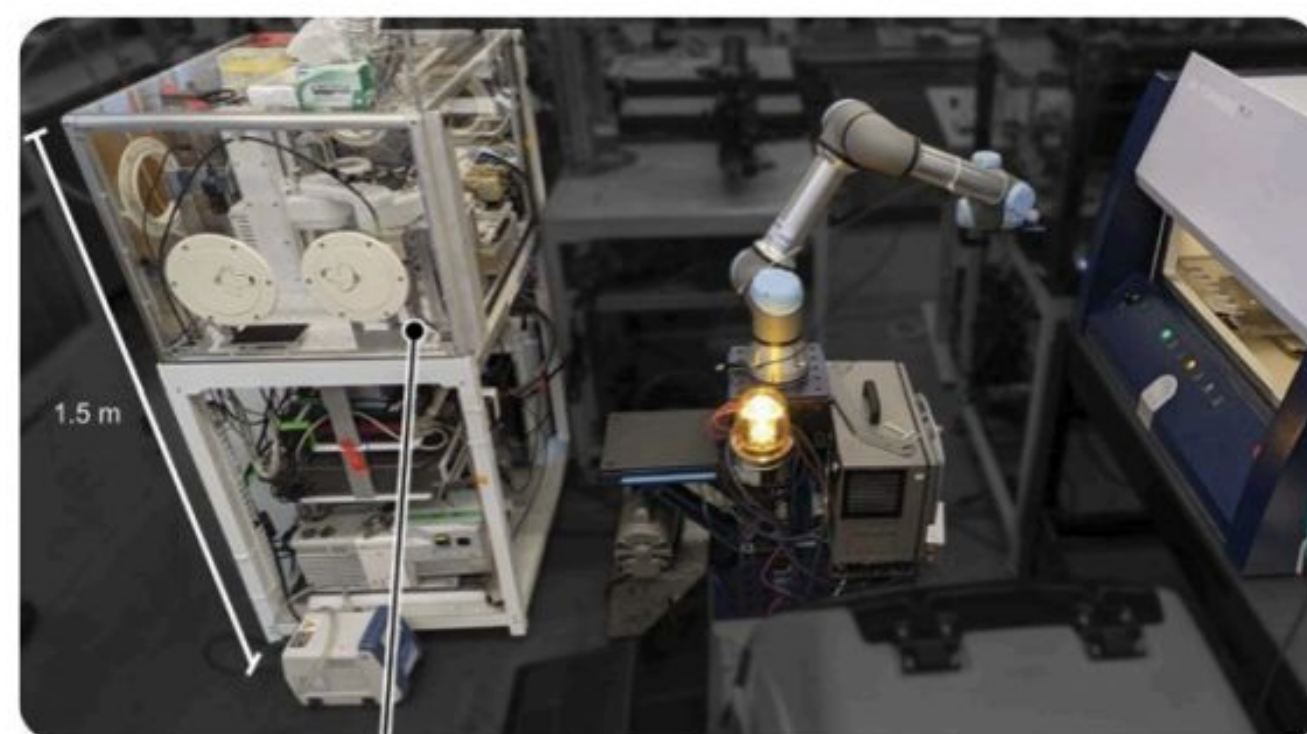
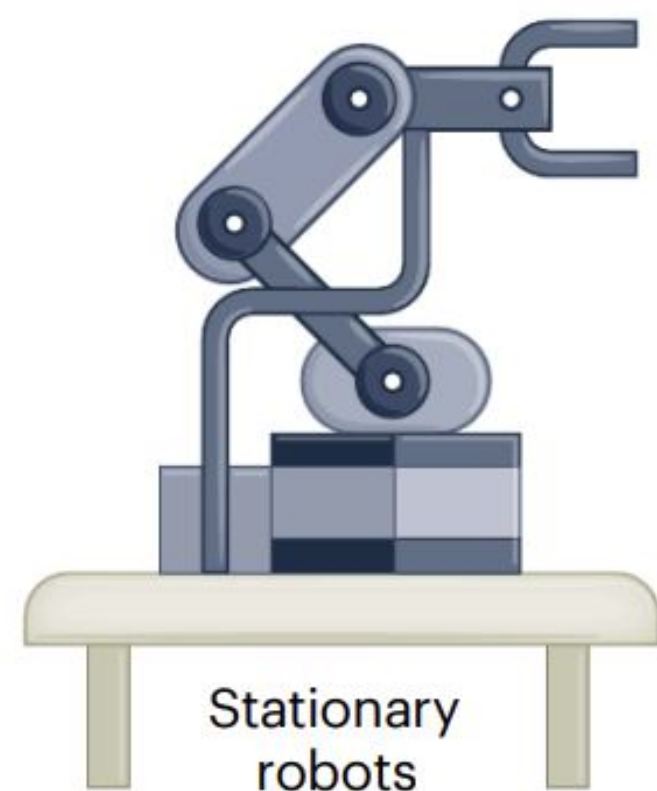
- Understand structure–property relationships to identify promising regions of the design space
- Predict properties, rank candidates
- Prioritize experimental validation



Bottleneck #1: Synthesis/Experimentation

How to accelerate experimental studies?

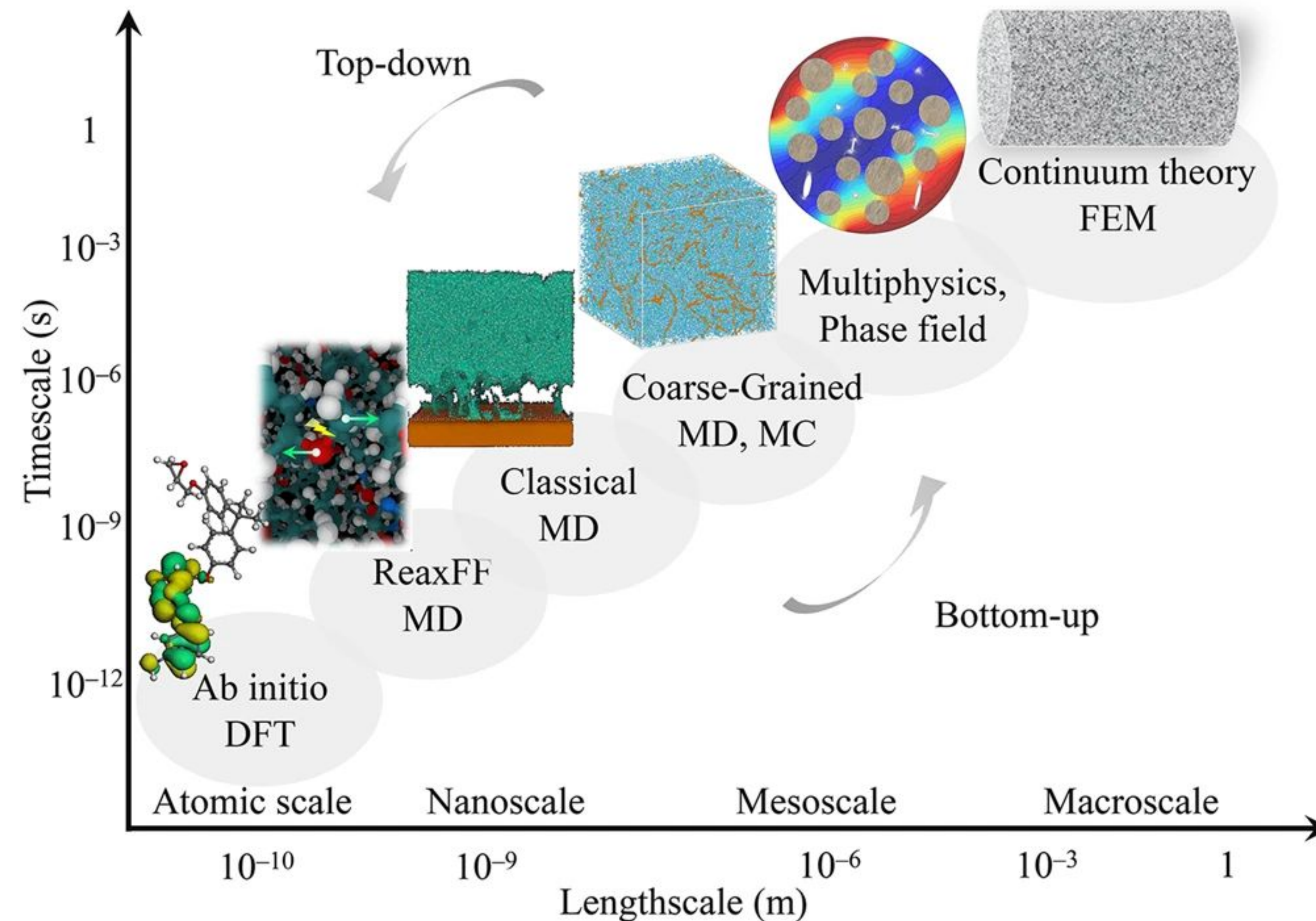
- Optimize synthesis protocols
- Automate experiments, characterization
- Accelerate post-processing routines



Bottleneck #2: Size of a search space

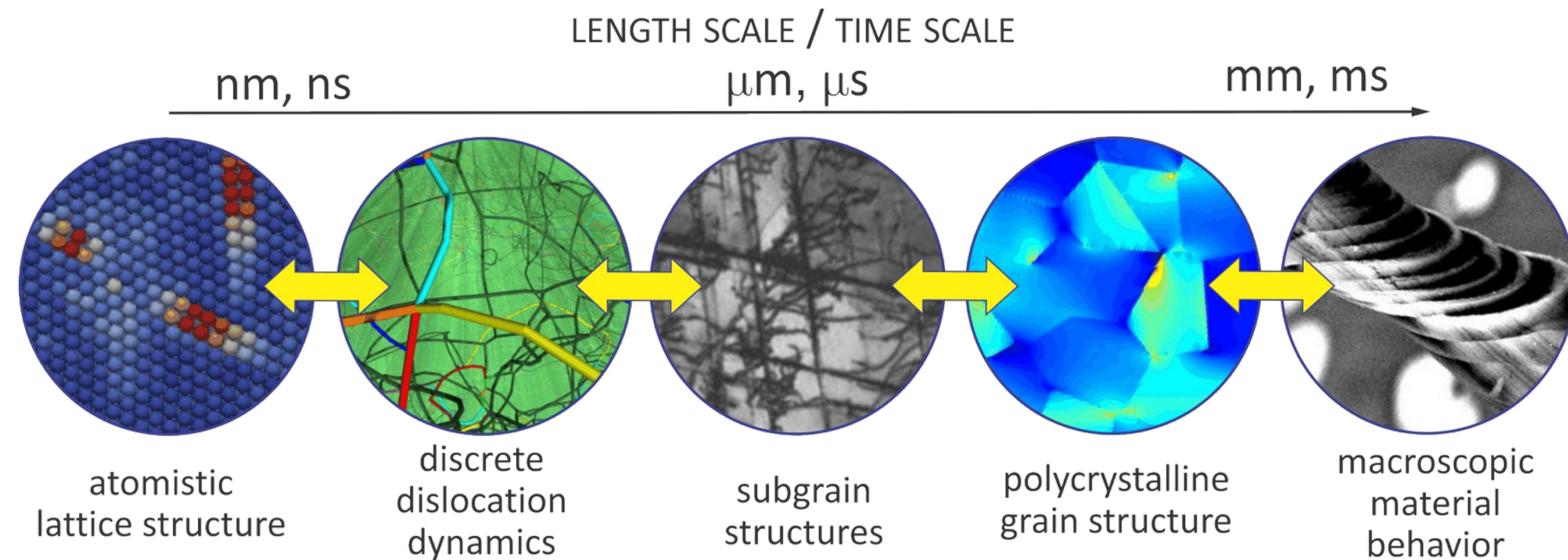
How to navigate this space before stepping in a lab?

- Understand structure–property relationships to identify promising regions of the design space
- Predict properties, rank candidates
- Prioritize experimental validation



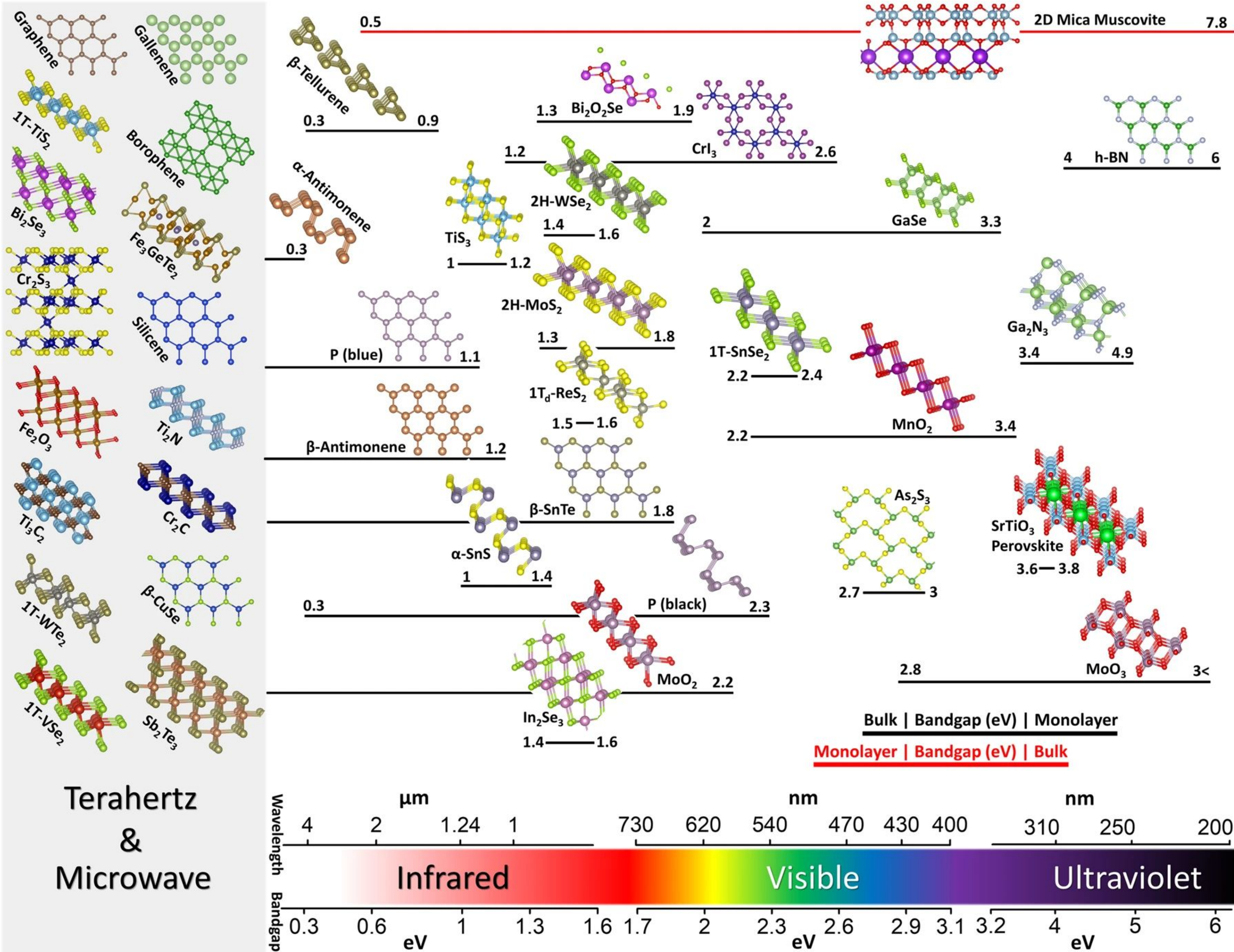
Materials are
characterized
...at various length
scales

- atomic
- micro
- meso
- macro



The course focuses
...on the atomic scale
...and inorganic materials

Bandgap engineering of two-dimensional semiconductor materials



Experiment ~~vs.~~ and atomistic modeling compromise

Experiment is the only source of truth, but:

- Expensive/Requires specific equipment
- Sequential
- Some phenomena cannot be observed directly

Atomistic modeling is approximation of reality, but:

- Atomic scale resolution
- Easily parallelizable
- Good agreement with exp./Interpretation of exp.

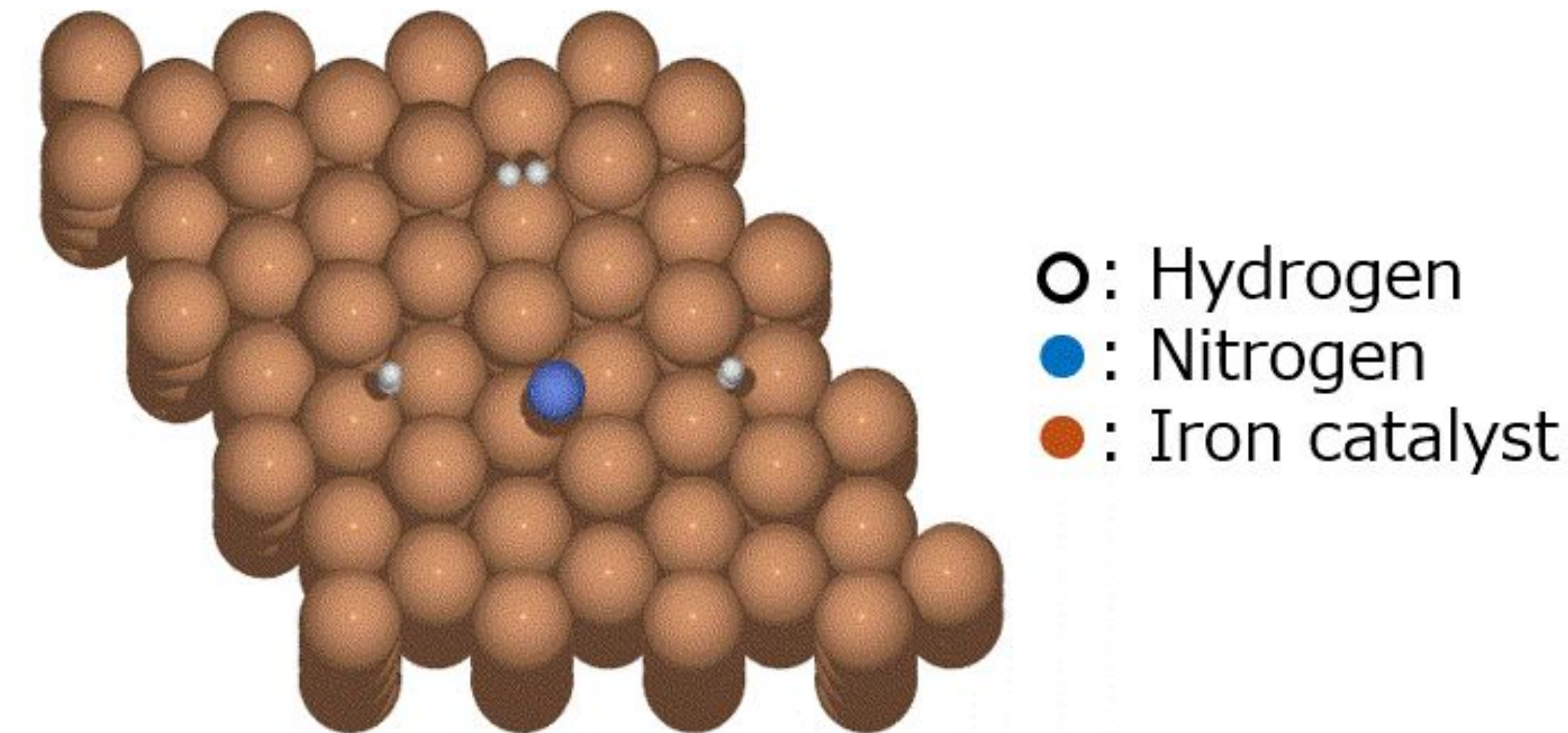
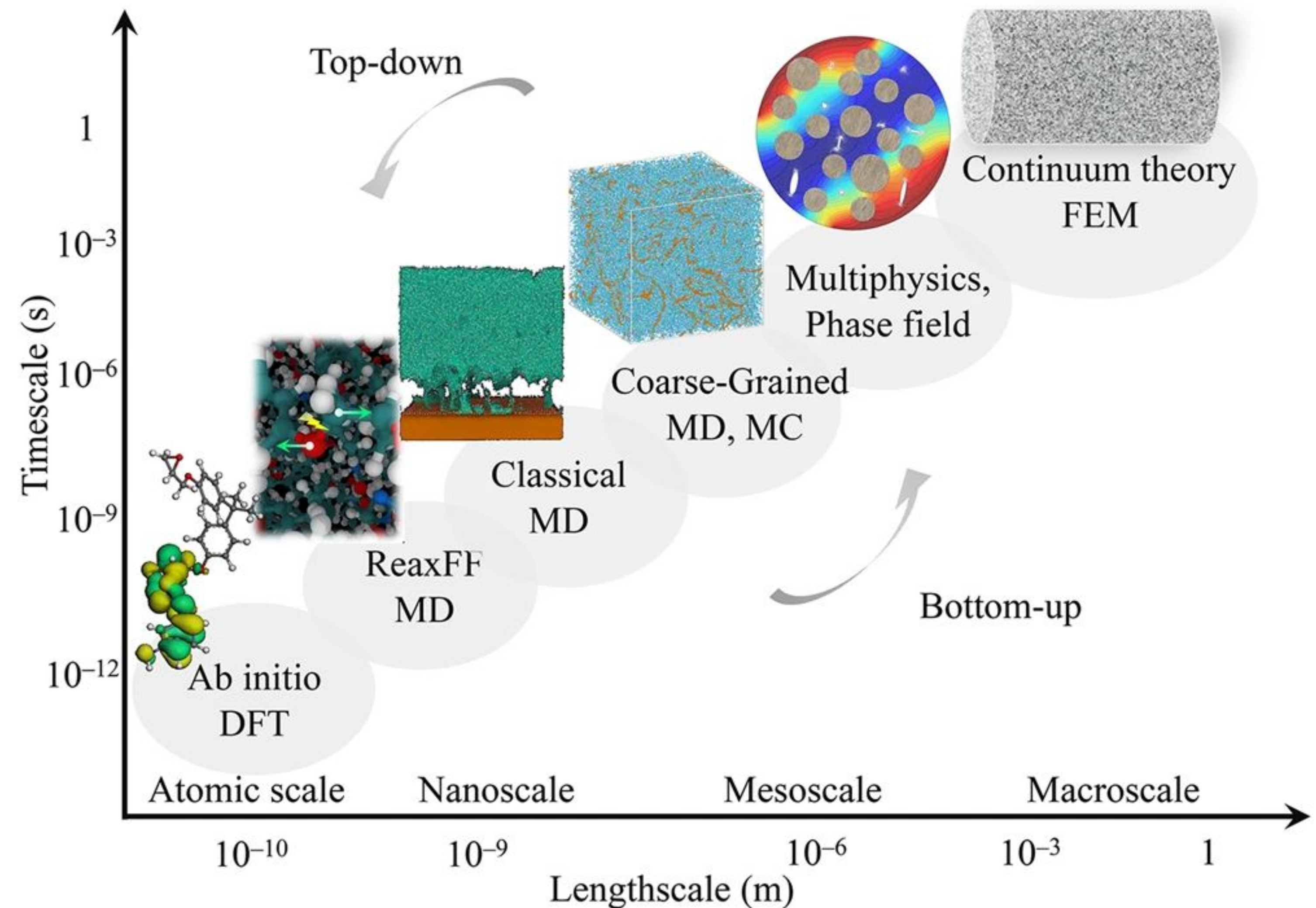


Image of ammonia synthesis

Trial-and-error loop can be optimized

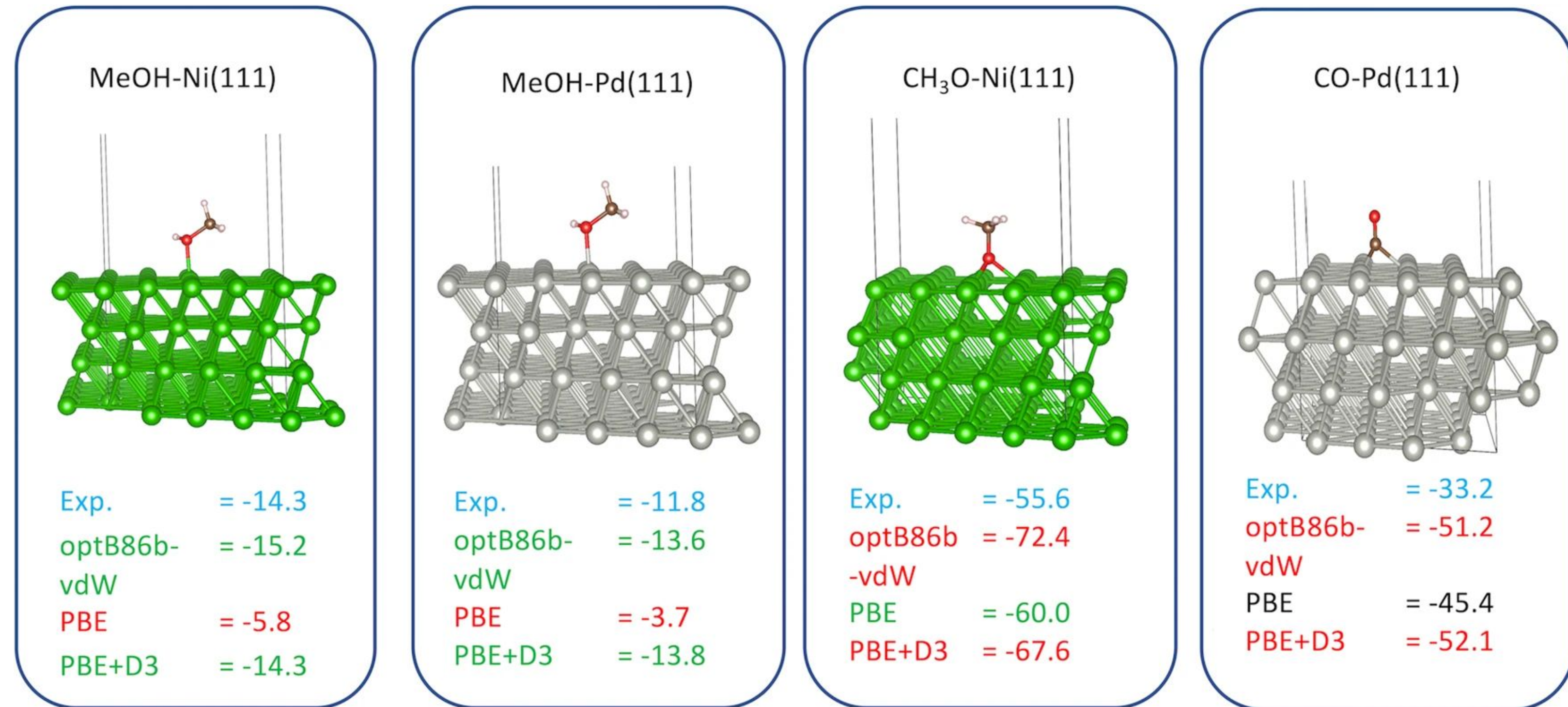
...with the help of theoretical and computational approaches

...by replacing “expensive” experiments with computation



Example: Adsorption energy

- Determines catalytic properties
 - Essential for chemical industry

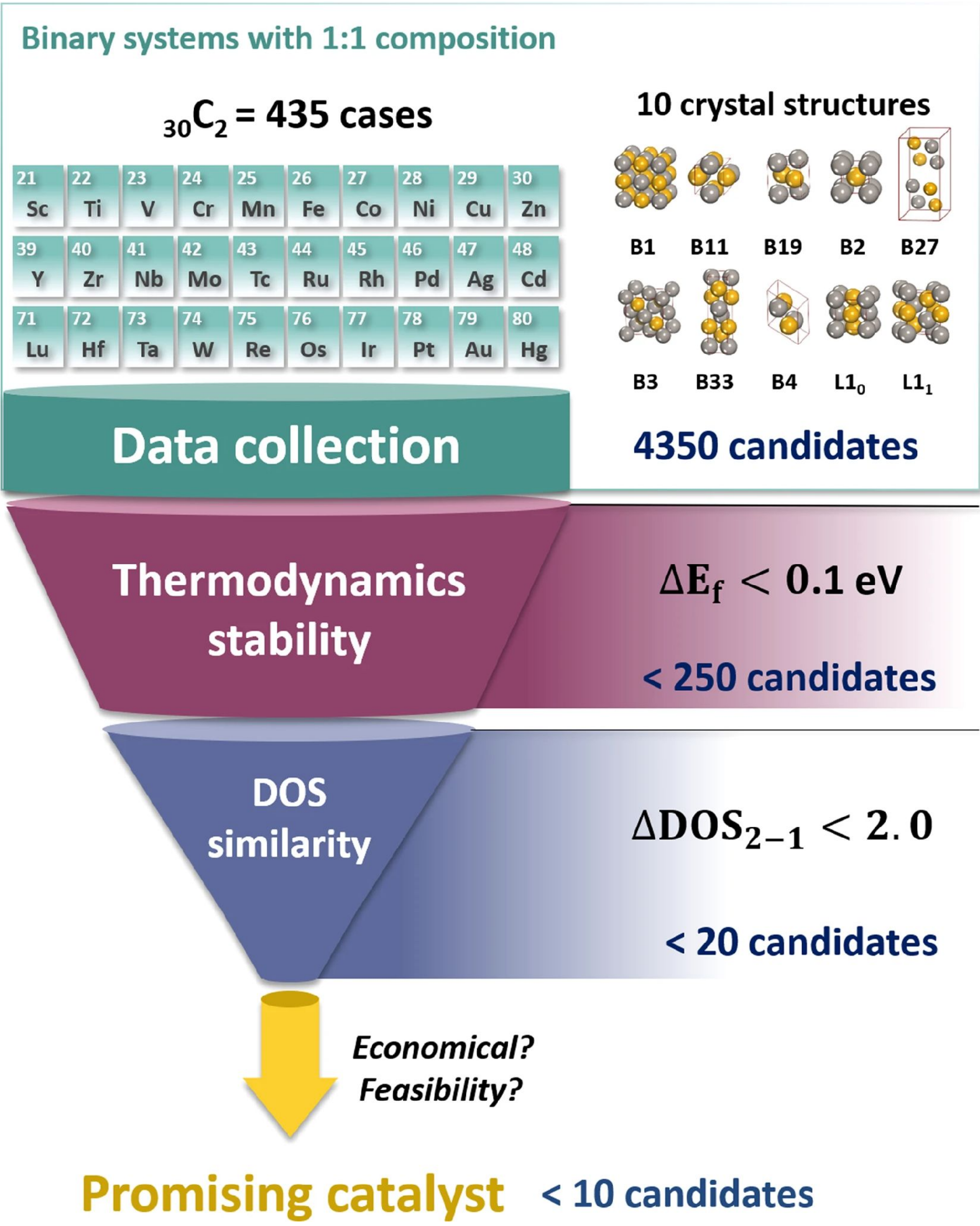


Adsorption energies on transition metal surfaces:
towards an accurate and balanced description
accurate and balanced description

Computational methods

...accelerate the first stage of the materials design

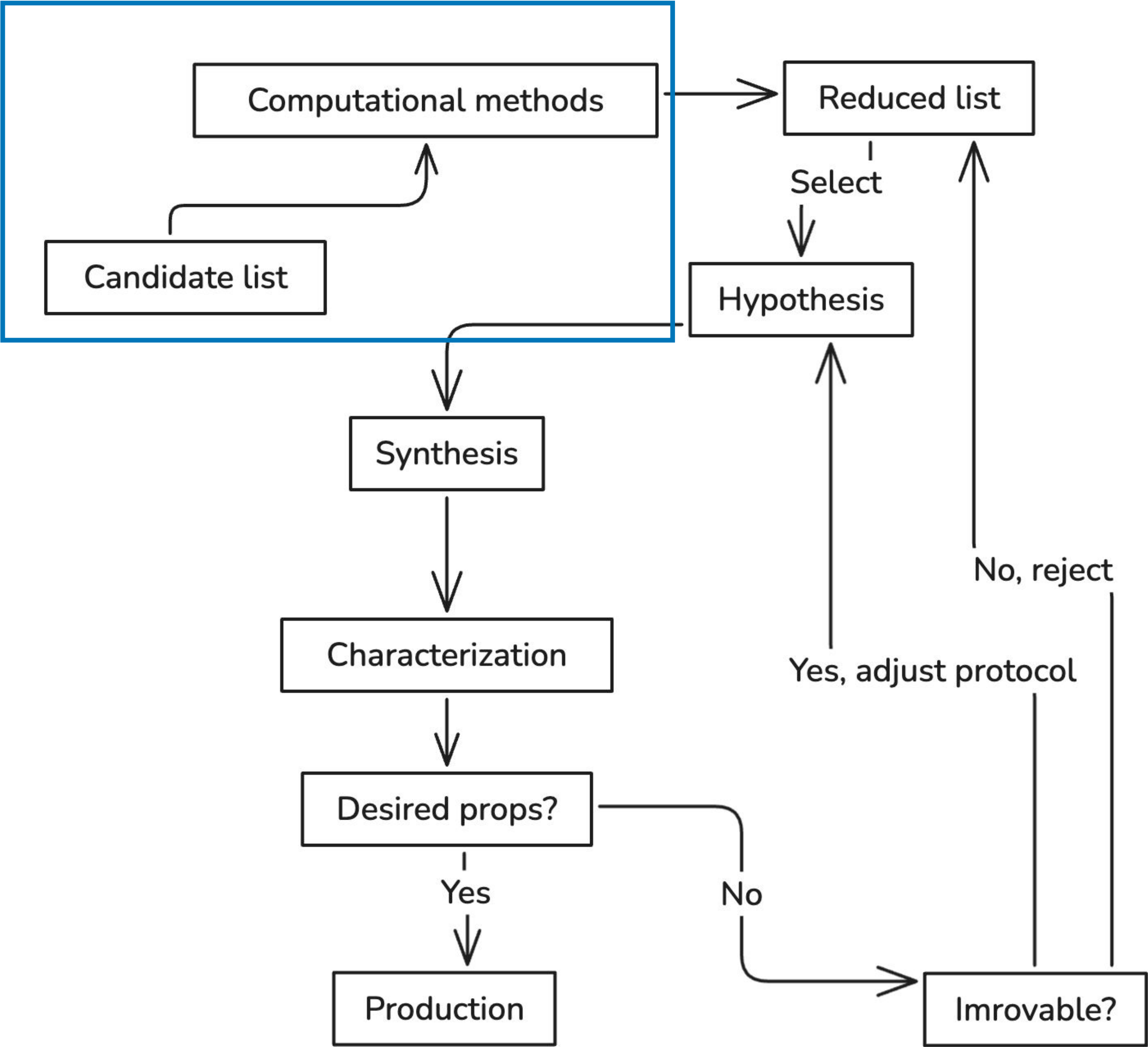
High-throughput computational-experimental screening protocol for the discovery of bimetallic catalysts **discovery of bimetallic catalysts**



Acceleration comes from here

Trial-and-error with computational methods

...takes ~2 years to design a material

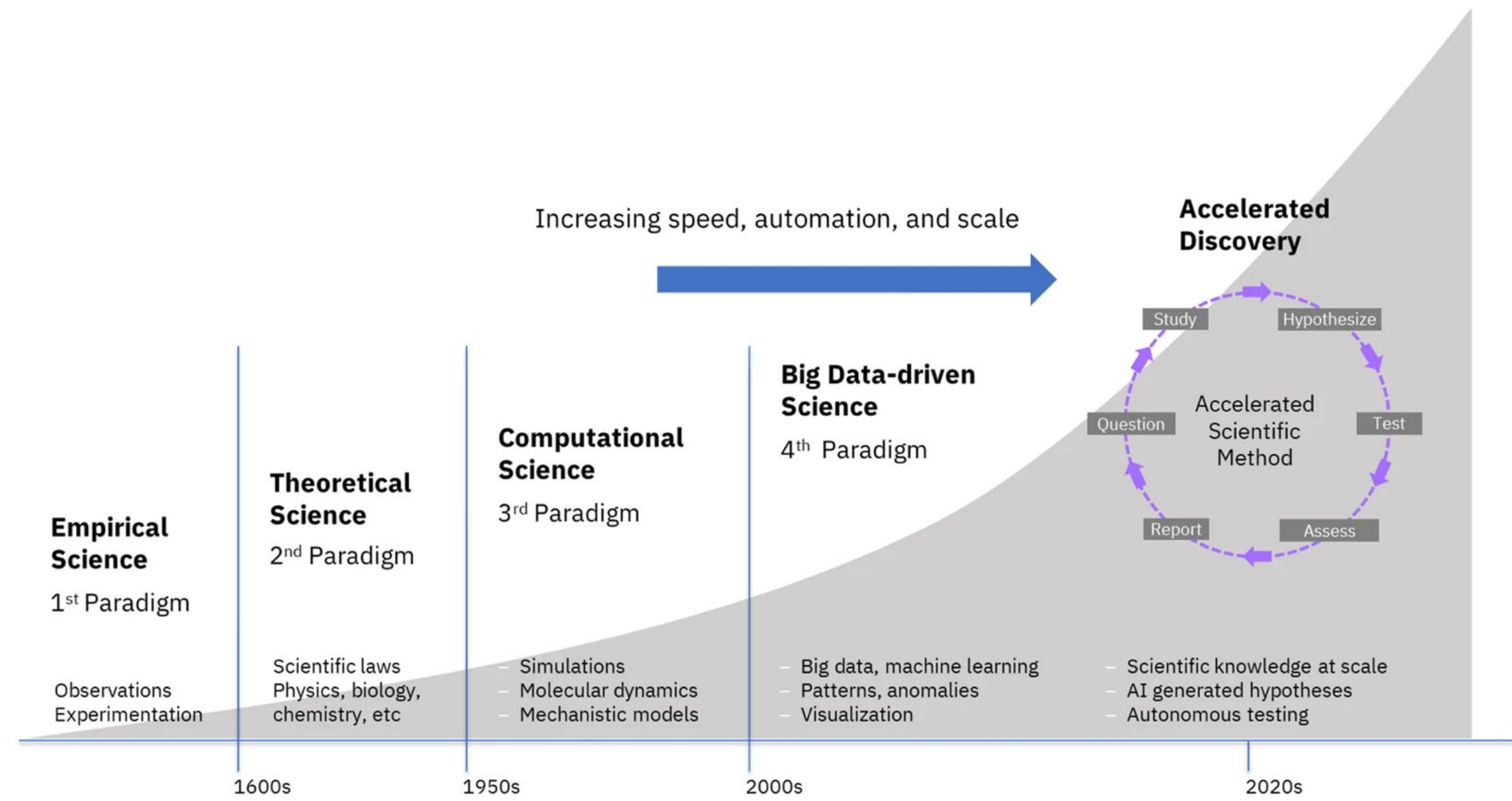


Can it be further optimized?

Materials informatics

...accelerates the solution of materials science problems through the use of data

...mostly by replacing computational methods with surrogate models

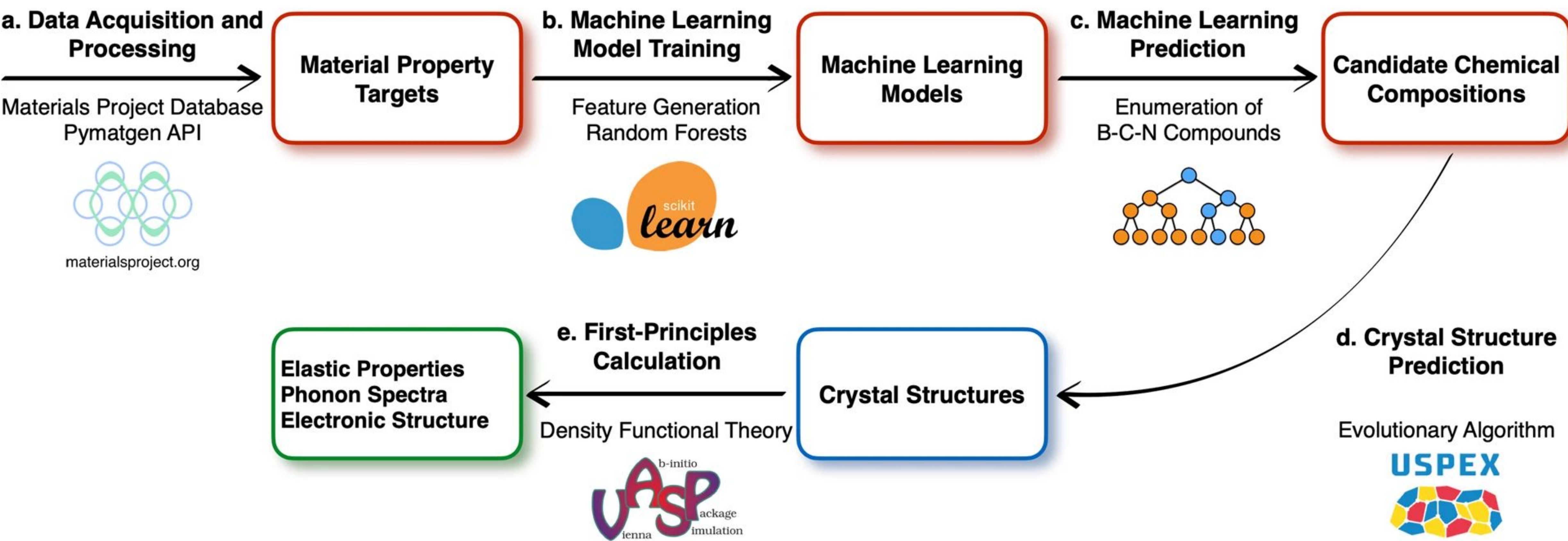


“A surrogate model is an engineering method used when an outcome of interest cannot be easily measured or computed, so an approximate mathematical model of the outcome is used instead.”

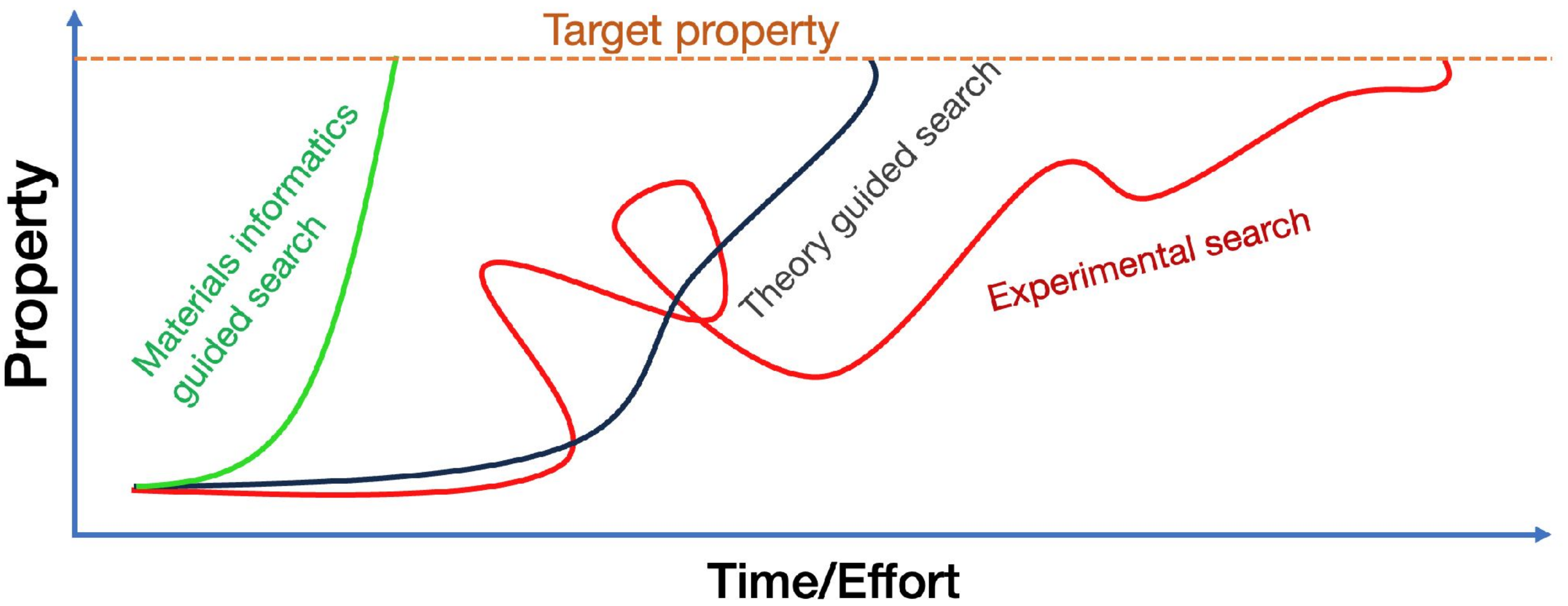
https://en.wikipedia.org/wiki/Surrogate_model

A typical materials informatics workflow

collect data -> train model -> predict -> reduce candidate list -> computational methods -> reduce candidate list -> experiment



All in all: Materials informatics paradigm for materials design



Part #2: Motivation, navigation, and assessment

Course structure

- 11 classes
- Each class = 1h lecture + 2h seminar
- 9 lectures
- 9 seminars
- 2 HWs (coding)
- 1 FP
- 1 Oral exam + HWs defense
- ~7 lecture/seminar recap quizzes

The blue text is clickable links

- Check this [link](#)

Class #1: What is materials informatics + Python crash course

Class #2: Python libraries for atomistic modelling of materials

Class #3: Data in materials science

Class #4: Data exploration, visualization, and fitting

Class #5: Classical ML for materials science pt.1

Class #6: Classical ML for materials science pt.2

Class #7: Graph neural networks for materials science pt.1

Class #8: Graph neural networks for materials science pt.2

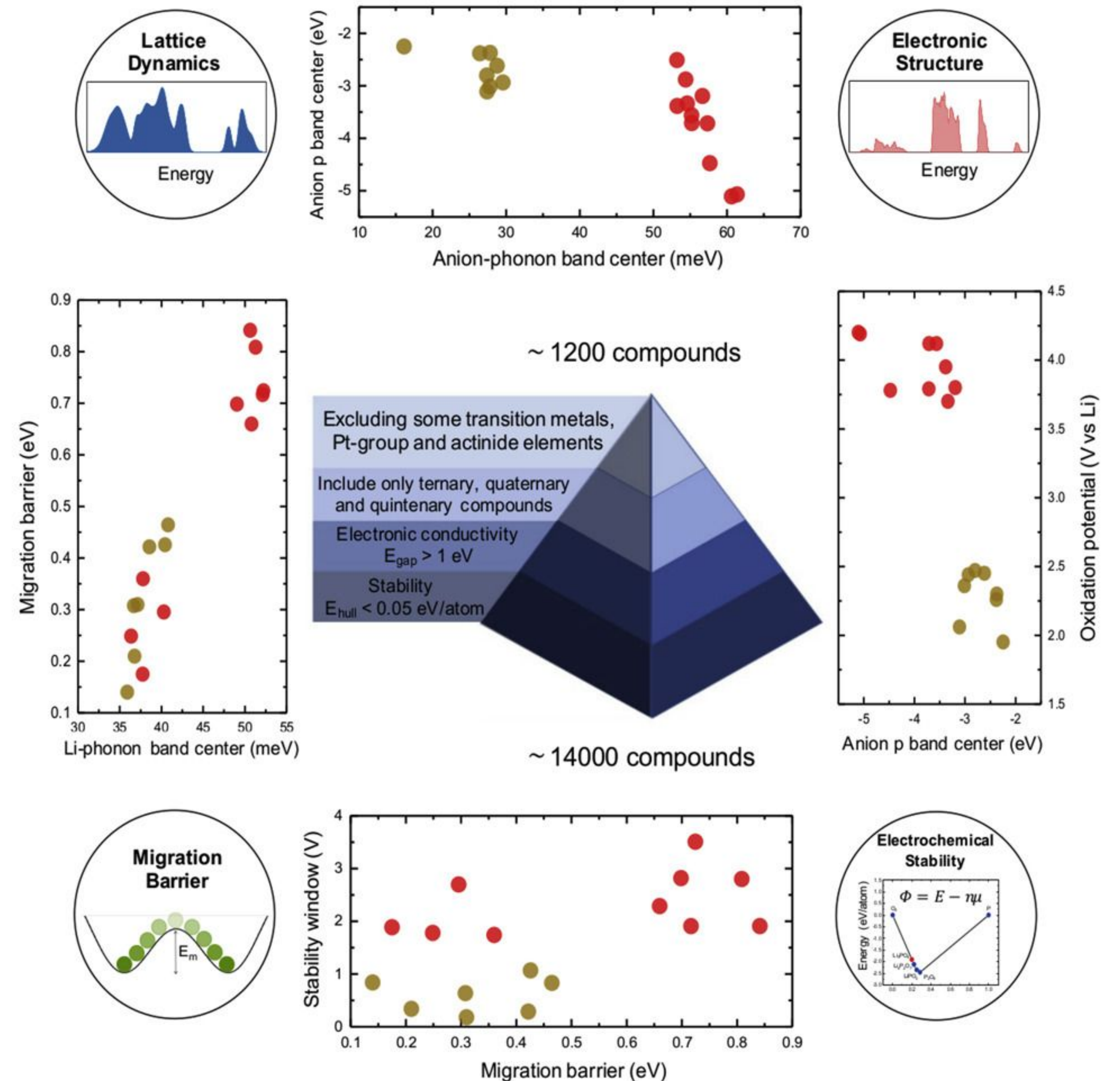
Class #9: Machine learning for molecular simulation

Class #10: Oral exam + HWs defense

Class #11: Final project presentations

FP example: High throughput screening of solid electrolytes for Li-ion batteries

- Formulate selection criteria
- Download the data from the database
- Prepare a dataset of ionic conductivity parameters
- Training: Select and train model/Interpret results (feature importance, etc.)
- Inference: Predict properties for candidate materials
- Perform a diffusion simulation study for the best candidate using universal interatomic potential
- Write a 3 page article style report
- Prepare a 5 minutes oral presentation



Your final project must be in line with the ILOs

On completion of the course you will be able to:

- Apply python libraries and data science tools to solve materials science problems
- Collect, generate and analyse materials science datasets, including identification of structure-property relationships
- Critically evaluate materials informatics literature

Course materials

- The course content is stored at the GitHub [repo](#)
- HWs are announced separately in the canvas and a telegram chat

Why?

Modern materials science = Materials informatics

Grading scheme

- Quizzes 10%
- HW 30%
- Oral exam 30%
- Final project 30%
 - Written report 50%
 - Oral presentation 40-50%
 - Discussion of other projects 0-10%

AI usage policies and control

- **AI tools are permitted.** Students may use AI-based tools, including large language models (LLMs), when solving their homework assignments.
- **Students are responsible for their submissions.** Students must understand and be able to explain every step of the solutions they submit, regardless of whether AI tools were used in producing them.
- **Verification during the oral exam.** The final evaluation of whether a student has achieved the intended learning outcomes at the required threshold will be conducted during the oral exam session. In addition to the regular exam questions, students may be asked to explain, justify, or elaborate on selected tasks and solutions from their submitted homework.

Take home message

- Materials informatics accelerates the solution of materials science problems through the use of data science tools
- We focus on atomistic scale and inorganic compounds
- By the end of the course you will be able to apply data science pipelines for solving materials science problems

Announcement

- HW1 is released
- HW1 deadline is Thursday, October 8th at 11.59 pm
- The deadline is strict
- Topics covered in HW1 (the first 4 lectures/seminars):
 - python for science: numpy, pandas, scipy, matplotlib
 - python for materials modeling: ase, pymatgen
 - python for materials informatics: The Materials project API, exploratory data analysis

Resources

<https://github.com/tilde-lab/awesome-materials-informatics>

<https://github.com/anthony-wang/BestPractices>

<https://github.com/sp8rks/MaterialsInformatics>

<https://github.com/ncfrey/resources>

<https://github.com/eddotman/intro-to-materials-informatics>

<https://enze-chen.github.io/mi-book-2021/intro.html>

Thank you for your attention!

Additional slides

Why are diamond and graphite so different despite the same chemical composition?

